

Hongkang Chu

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Multimodal learning; AI agents; synthetic image and video detection; MRI and MRSI reconstruction

Education

University of the Chinese Academy of Sciences Master's student, Computer Science. Expected graduation: 2028.	2025–present
Shanghai Ocean University Bachelor's degree, Computer Science and Technology. Undergraduate research in protein language models.	2021–2025

Research Experience

Chinese Academy of Sciences / UCAS <i>Graduate Research</i> Research in MRI, MRSI, and deuterium metabolic imaging at the Innovation Academy for Precision Measurement Science and Technology.	2025–present
Zhejiang University <i>Research Collaboration</i> Collaborative research in detecting AI-generated images and videos.	Apr–Sep 2026
University of Illinois Urbana-Champaign <i>Remote Research Intern</i> Research in multimodal models and medical AI.	Feb 2026–present
Shanghai Ocean University <i>Undergraduate Research</i> Protein language models for druggable protein classification.	2021–2025

Selected Publications and Conference Contributions

TRACE: Trajectory Representation and Consistency Estimation for AI-Generated Video Detection <i>ICLR 2027</i> . Under Review. Second Author.
ClueAegis: Heuristic-to-Reasoning Cognitive-skill Learning for Unified Evidence-based Synthetic Image Detection <i>Findings of EMNLP 2026</i> . Accepted. Second Author.
Physics-embedded CycleGAN for robust metabolite mapping in short-readout deuterium metabolic imaging <i>Magnetic Resonance Letters, 2026</i> . Published Online. Co-author.
Phase Model-Driven Deep Learning for Robust Phase Correction in High-Throughput NMR-Based Metabolomics <i>The Journal of Physical Chemistry Letters, 2026</i> . Published. Co-author.
Deep Learning Enhanced Weighted Average CSI for High-Resolution Deuterium Metabolic Imaging <i>ISMRM 2026</i> . Power Pitch Oral. First Author.
Comprehensive Research on Druggable Proteins: From PSSM to Pre-Trained Language Models <i>International Journal of Molecular Sciences, 2024</i> . Published. First Author.